

CHAPTER 1: FOUNDRY TECHNOLOGY

1. Six Applications of Casting (3 Marks): Easy Win (Appeared in actual mid-sem). Write these exact 6 points: 1. Automotive engine blocks and cylinder heads. 2. Machine tool beds and frames (due to damping capacity). 3. Turbine blades (aerospace). 4. Railway wheels and tracks. 5. Pump housings and valve bodies. 6. Agricultural machinery components.

2. Pattern Allowances - Theory (7 Marks): Tier 1. What to write for full marks: • Definition: Intentional dimensional modifications made to a pattern to compensate for metal shrinkage, machining, and casting defects. • 1. Shrinkage Allowance: Metals shrink when cooling from liquid to solid. Pattern is made larger. (Values: CI = 10mm/m, Steel = 20mm/m, Al = 13mm/m). • 2. Machining/Finish Allowance: Extra material added to surfaces that require machining later. (Usually 1.5 to 6 mm). • 3. Draft Allowance: Taper applied to vertical faces of the pattern for easy withdrawal without damaging the mould. (1° to 3° external, 2° to 5° internal). • 4. Shake/Rapping Allowance: When a pattern is shaken to remove it, the mould cavity enlarges. Pattern is made slightly smaller to compensate. (Negative allowance). • 5. Distortion/Camber Allowance: Applied to U or V-shaped castings that tend to warp during cooling. Pattern is bent in the opposite direction. • Diagram Required (Crucial for 7M): Draw a 2D cross-section of a solid block. Show a dotted line outside for "Machining Allowance" and angled vertical sides for "Draft Allowance".

3. Pattern Allowance - Numerical (7 Marks): Tier 1 (From actual mid-sem). How to score full marks: 1. Read dimensions: Identify L, W, H of the final casting. 2. Add Shrinkage: Add the % or mm/m shrinkage to each dimension. Formula: $\text{Dimension}_{\text{new}} = \text{Dimension}_{\text{old}} + (\text{Dimension}_{\text{old}} \times \frac{\text{Shrinkage}\%}{100})$. 3. Add Machining: Add machining allowance to every face that needs it. Rule: For a cube, machining of 2mm/side means adding 4mm to length, 4mm to width, 4mm to height (2 sides per dimension). 4. Add Draft: Applied to vertical dimensions. Add the taper value.

4. Types of Patterns (3-4 Marks): Tier 1. List these and draw rough sketches for 3 of them: 1. Single piece / Solid pattern (Sketch: Simple block). 2. Split / Two-piece pattern (Sketch: Two halves aligned with dowel pins). 3. Match plate pattern (Sketch: Central plate with pattern halves mounted on both sides). 4. Cope and drag pattern. 5. Gated pattern (Sketch: Multiple patterns connected by a central runner/gate). 6. Sweep pattern (Sketch: Central vertical spindle with a sweeping wooden board). 7. Skeleton pattern. 8. Loose piece pattern.

5. Investment Casting (7 Marks): Tier 1. What to write for full marks: 1. Principle: Uses a wax pattern destroyed during the process to create a seamless cavity (also called lost-wax process). 2. Steps (must list in order): • Wax pattern creation. • Assembly into a tree. • Slurry coating (dipping in refractory ceramic). • Stuccoing (sanding) and drying to form a shell. • Dewaxing (heating to melt wax out). • Pouring molten metal. • Knockout (breaking ceramic shell). 3. Diagram: Draw three stages: (1) The wax tree, (2) The dipping process, (3) The final pouring of metal into the hollow ceramic shell. 4. Advantages: Excellent surface finish, extremely high dimensional accuracy, no parting line, can cast complex shapes. 5. Limitations (Must include for 7M): High cost, process is slow, limited to smaller castings (weight restriction).

CHAPTER 2: METAL JOINING

6. Gas Welding & 3 Flames (7 Marks): Tier 1 - Highest Probability. What to write for full marks: • Definition: Joining metals by melting them using an oxy-acetylene gas flame. • Equipment List (Add for 7M): Oxygen cylinder (black), Acetylene cylinder (maroon), pressure regulators, hoses, welding torch. • Flames (MUST draw all 3): 1. Neutral Flame (Equal O₂ and C₂H₂): Diagram: Draw two zones: Inner luminous cone, outer envelope. Temp: ~3200°C. Use: Mild steel, cast iron, copper, aluminum. 2. Oxidizing Flame (Excess O₂): Diagram: Draw two zones: Short, pointed inner cone, shorter outer envelope. Temp: ~3300°C. Use: Brass, bronze (copper-base alloys). 3. Carburizing/Reducing Flame (Excess C₂H₂): Diagram: Draw three zones: Inner cone, intermediate acetylene feather, outer envelope. Temp: ~3100°C. Use: High carbon steels, hard-facing.

7. Electrode Coating Purpose (3 Marks): Easy Win. List these 4 exact points: 1. Provides a gaseous shield to protect molten weld pool from atmospheric oxygen and nitrogen. 2. Forms slag over the weld to protect it during cooling and refine the metal. 3. Stabilizes the arc (contains ionizing agents). 4. Adds alloying elements to the weld metal to improve properties.

8. Polarity in Arc Welding (3 Marks): Easy Win. What to write: • DCSP (Direct Current Straight Polarity): Workpiece is Positive (+), Electrode is Negative (-). 66% of heat is at the workpiece. Used for thick plates. • DCRP (Direct Current Reverse Polarity): Workpiece is Negative (-), Electrode is Positive (+). 66% of heat is at the electrode. Used for thin plates (prevents burn-through) and metals with refractory oxides (Aluminum).

9. Brazing vs Soldering (3-4 Marks): Easy Win. • Temperature: Brazing is above 450°C (but below base metal melting point), Soldering is below 450°C. • Filler Metal: Brazing uses Spelter (Copper, Zinc, Silver alloys), Soldering uses Solder (Lead-Tin alloys). • Strength: Brazing has a stronger joint, handles higher stress; Soldering has a weaker joint, handles low stress. • Application: Brazing for pipe fittings, carbide tool tips; Soldering for electronics, PCB boards, plumbing.

CHAPTER 3: FORMING & SHAPING

10. Hot vs Cold Working (4-7 Marks): Tier 1 & Easy Win. • Temperature: Hot Working is above recrystallization temperature; Cold Working is below. • Force Required: Hot Working requires less force/power; Cold Working requires high force/power. • Surface Finish: Hot Working has poor surface finish (oxidation/scale); Cold Working has excellent surface finish. • Grain Structure: Hot Working forms new equiaxed grains; Cold Working elongates (distorts) grains. • Strength/Hardness: Hot Working remains unchanged; Cold Working increases due to strain hardening. • Residual Stress: Hot Working is negligible; Cold Working has high residual stresses.

11. Rolling: Derivation $l = \sqrt{R \Delta t}$ (7 Marks): Tier 1 (From actual mid-sem). Steps for full marks: 1. Diagram: Draw two rolls (radius R), plate entering (thickness t_0) and leaving (thickness t_f). Show the contact angle α and projected contact length l . 2. Draft: Define draft $\Delta t = t_0 - t_f$. 3. Geometry: Look at the triangle formed by the roll center, the entry point, and the vertical axis. 4. The vertical leg is $R - (\Delta t / 2)$. 5. The horizontal leg is l . 6. The hypotenuse is R . 7. Pythagoras: $R^2 = l^2 + (R - \frac{\Delta t}{2})^2$. 8. Expand: $R^2 = l^2 + R^2 - R \Delta t + \frac{\Delta t^2}{4}$. 9. Cancel R^2 . Assume $\frac{\Delta t^2}{4}$ is very small and ignore it. 10. Result: $0 = l^2 - R \Delta t$ implies $l^2 = R \Delta t$ implies $l = \sqrt{R \Delta t}$.

12. Rolling Numerical Formulas (7 Marks): Tier 1 (From actual mid-sem). Memorize these to solve the 7-mark torque/power question: 1. Draft: $\Delta t = h_0 - h_1$. 2. Contact Length: $L_c = \sqrt{R \cdot \Delta t}$. 3. True Strain: $\epsilon = \ln(\frac{h_0}{h_1})$. 4. Average Flow Stress ($\bar{Y}_f$): If given directly, use it. (If given as $K \epsilon^n$, then $\bar{Y}_f = \frac{K \epsilon^{n+1}}{n+1}$). 5. Roll Force (F): $F = \bar{Y}_f \cdot w \cdot L_c$ (where w = width of plate). 6. Torque (T): $T = F \cdot a$ (where a is the moment arm. Usually $a = 0.5 \cdot L_c$ for hot rolling). For ONE roll. 7. Total Power (P): $P = 2 \cdot \frac{\pi N T}{60}$ (Multiply by 2 because there are two rolls).

13. Extrusion: Forward vs Backward (7 Marks): Tier 1. What to write for full marks: • Forward (Direct) Extrusion: Billet is placed in a chamber. Ram pushes the billet. The metal flows in the same direction as the ram movement. Sketch: Show ram pushing right, metal exiting right. Advantage: Simpler tooling. Disadvantage: High friction between billet and chamber walls. High force required. High scrap (butt). • Backward (Indirect) Extrusion: Die is mounted on a hollow ram. Ram pushes into the billet. Metal flows in the opposite direction to the ram movement through the hollow ram. Sketch: Show ram pushing right, metal exiting left (backwards) through the hollow ram. Advantage: Zero wall friction. 25-30% less force required than forward extrusion. Disadvantage: Mechanically complex tooling, hollow ram limits the maximum load that can be applied.

14. Wire vs Tube Drawing (4-7 Marks): Tier 1. • Wire Drawing: Pulling a solid metal rod through a converging die to reduce its diameter. Force is applied as tension at the exit. • Tube Drawing: Reducing the diameter and wall thickness of a hollow tube. • Key Difference for Full Marks: Tube drawing requires a mandrel inside the tube to support the internal walls and control the final internal diameter. Wire drawing has no internal support. • Diagrams to Draw: 1. Wire drawing die (solid metal passing through). 2. Tube drawing setup showing the die AND the Mandrel. • Types of Mandrels (Mention for 7M): Fixed mandrel, floating mandrel, moving mandrel.